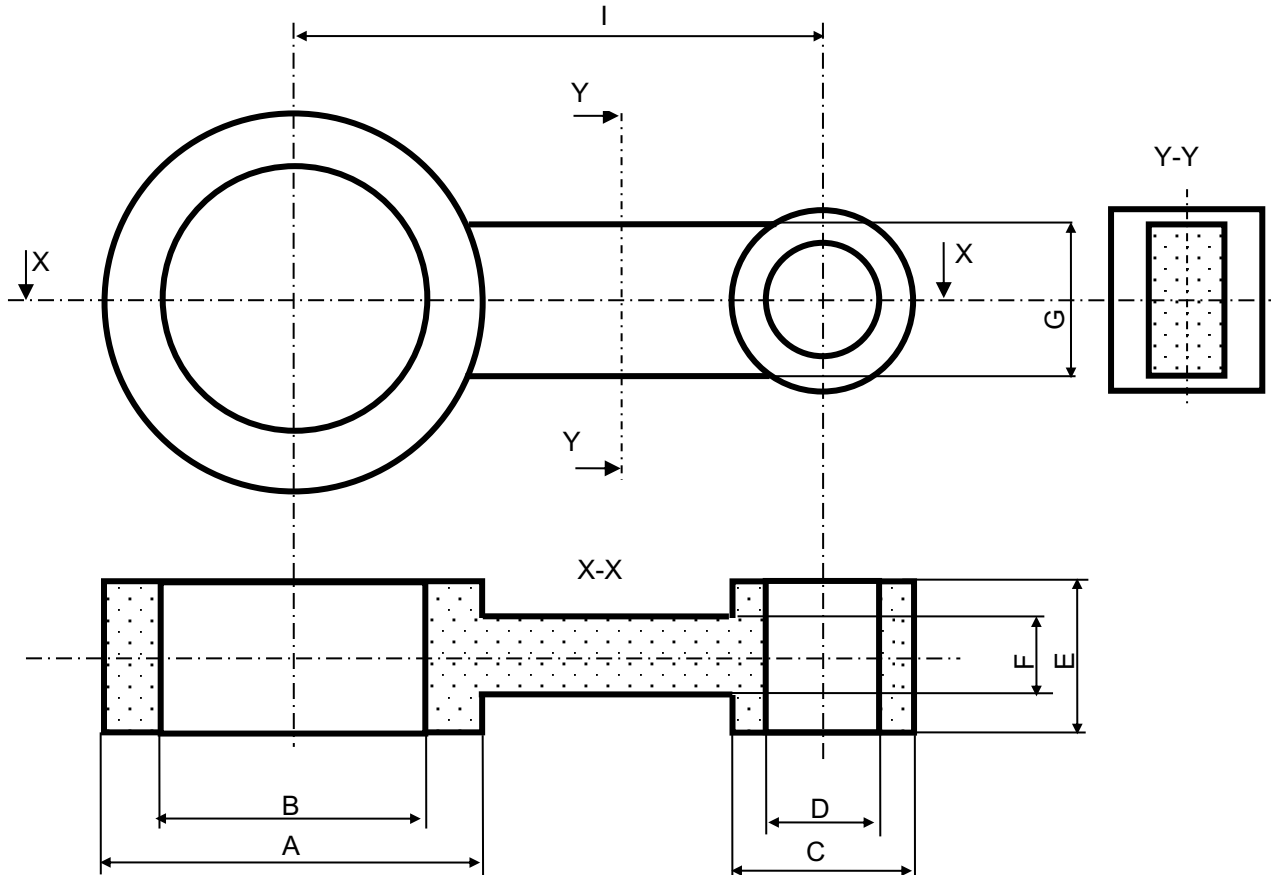


SOLUTIONS TO NUMERICAL PROBLEMS

METAL CASTING



The figure shows a casting to be fabricated by an expendable mould process. The casting will be made of steel and will have the following final dimensions after machining: $A = 50$ mm, $B = 35$ mm, $C = 24$ mm, $D = 15$ mm, $E = 20$ mm, $F = 10$ mm, $G = 20$ mm, $I = 70$ mm.

- Design the pattern and any cores respecting a machining allowance equal to 2 mm and a shrinkage allowance equal to 1,4% (for simplicity, draft angles and fillets/radii are omitted).
- Considering the original dimensions, study the directional solidification of the metal. Is the directional solidification feasible? (Consider three elements: Element A is the cylinder of external diameter A; Element C is cylinder of external diameter C, Element B is the middle block to be considered tangent to the cylinders)
- Considering: the bottom plane of the casting as parting line, $A_s:A_r:A_g = 4:3:2$, $c = 0,5$, a volume of the casting equal to 36 cm^3 , a top riser with a volume equal to 15 cm^3 , a cope height equal to 200 mm. Design a bottom gating system so that the mould filling time is equal to 12 s.

Solution

- Design the pattern and any cores respecting a machining allowance equal to 2 mm and a shrinkage allowance equal to 1,4% (for simplicity, draft angles and fillets/radii are omitted).

In order to find the pattern dimensions, we need to consider the material allowance needed for the following machining operations and the shrinkage allowance (1,4%). The result must be rounded to mm: upper integer for external dimensions, lower integer for internal dimensions. Therefore (for sake of completeness the computation is done for all the pattern dimensions):

$$\begin{aligned}
A &= (50 + 2 \cdot 2)(1 + 0,014) = 54 \cdot 1,014 = 54,756 = 55 \text{ mm} \\
B &= (35 - 2 \cdot 2)(1 + 0,014) = 31 \cdot 1,014 = 31,434 = 31 \text{ mm} \\
C &= (24 + 2 \cdot 2)(1 + 0,014) = 28 \cdot 1,014 = 28,392 = 29 \text{ mm} \\
D &= (15 - 2 \cdot 2)(1 + 0,014) = 11 \cdot 1,014 = 11,154 = 11 \text{ mm} \\
E &= (20 + 2 + 2)(1 + 0,014) = 24 \cdot 1,014 = 24,336 = 25 \text{ mm} \\
F &= (10 + 2 + 2)(1 + 0,014) = 14 \cdot 1,014 = 14,196 = 15 \text{ mm} \\
G &= (20 + 2 + 2)(1 + 0,014) = 24 \cdot 1,014 = 24,336 = 25 \text{ mm} \\
I &= 70 \cdot (1 + 0,014) = 70 \cdot 1,014 = 70,98 = 71 \text{ mm}
\end{aligned}$$

b. Considering the original dimensions, study the directional solidification of the metal. Is the directional solidification feasible?

Element A:

$$\begin{aligned}
V_A &= \frac{\pi}{4}(A^2 - B^2) \cdot E = \frac{\pi}{4}(50^2 - 35^2) \cdot 20 = 20.028 \text{ mm}^3 \\
S_A &= 2 \cdot \frac{\pi}{4}(A^2 - B^2) + \pi(A + B) \cdot E - F \cdot G = 2 \cdot \frac{\pi}{4}(50^2 - 35^2) + \pi(50 + 35) \cdot 20 - 10 \cdot 20 = 7.143 \text{ mm}^2 \\
M_A &= \frac{V_A}{S_A} = \frac{20.028}{7.143} = 2,80 \text{ mm}
\end{aligned}$$

Element C:

$$\begin{aligned}
V_C &= \frac{\pi}{4}(C^2 - D^2) \cdot E = \frac{\pi}{4}(24^2 - 15^2) \cdot 20 = 5.513 \text{ mm}^3 \\
S_C &= 2 \cdot \frac{\pi}{4}(C^2 - D^2) + \pi(C + D) \cdot E - F \cdot G = 2 \cdot \frac{\pi}{4}(24^2 - 15^2) + \pi(24 + 15) \cdot 20 - 10 \cdot 20 = 2.802 \text{ mm}^2 \\
M_C &= \frac{V_C}{S_C} = \frac{5.513}{2.802} = 1,97 \text{ mm}
\end{aligned}$$

Element B:

$$\begin{aligned}
L_B &= I - \frac{A}{2} - \frac{C}{2} = 70 - \frac{50}{2} - \frac{24}{2} = 33 \text{ mm} \\
V_B &= F \cdot G \cdot L_B = 10 \cdot 20 \cdot 33 = 6.600 \text{ mm}^3 \\
S_B &= 2 \cdot F \cdot L_B + 2 \cdot G \cdot L_B = 2L_B(F + G) = 2 \cdot 33(10 + 20) = 1.980 \text{ mm}^2 \\
M_B &= \frac{V_B}{S_B} = \frac{6.600}{1.980} = 3,33 \text{ mm}
\end{aligned}$$

Directional solidification:

The first element that will solidify is Element C, then Element A and at last Element B. Being Element B in between Elements A and C, the riser has to be positioned on the Element B (last to solidify).

Comparing the V/A ratios, it results:

$$\frac{M_B}{M_A} = \frac{3,33}{2,80} = 1,19 \in [1,1 \div 1,3] \rightarrow \text{Element B will correctly feed Element A}$$

$$\frac{M_B}{M_C} = \frac{3,33}{1,97} = 1,69 \in [1,1 \div 1,3] \rightarrow \text{Element B will feed Element C, but the ratio is too high}$$

Therefore, a change is needed to have a feasible casting process.

c. Considering: the bottom plane of the casting as parting line, As:Ar:Ag = 4:3:2, c = 0,5, a volume of the casting equal to 36 cm³, a top riser with a volume equal to 15 cm³, a cope height equal to 200 mm. Design a bottom gating system so that the mould filling time is equal to 12 s.

In the problem, a bottom gating system is used in order to fill a mould with a top riser. Therefore:

$$h_i = 200 \text{ mm}$$

$$h_f = 0 \text{ mm}$$

$$H_m = \left(\frac{\sqrt{h_i} + \sqrt{h_f}}{2} \right)^2 = \left(\frac{\sqrt{200} + \sqrt{0}}{2} \right)^2 = 50 \text{ mm}$$

Therefore, the pouring velocity in the bottleneck cross section (area of the gates, A_g) will be

$$v = c \cdot \sqrt{2 \cdot g \cdot H_m} = 0,5 \cdot \sqrt{2 \cdot 9,81 \cdot \frac{50}{1000}} = 0,495 \frac{\text{m}}{\text{s}} = 495 \frac{\text{mm}}{\text{s}}$$

Being the mould filling time equal to $T_{MF} = \frac{V_{total}}{A_{bottleneck} \cdot v}$, we get:

$$V_{total} = V_{casting} + V_{risers} = 36 + 15 = 51 \text{ cm}^3 = 51.000 \text{ mm}^3$$

$$A_{bottleneck} = \frac{V_{total}}{T_{MF} \cdot v} = \frac{51.000}{12 \cdot 495} = 8,59 \text{ mm}^2$$

Therefore, the design of the gating system will be characterized by the following areas of the gates, runners and downsprue:

$$A_g = A_{bottleneck} = 8,59 \text{ mm}^2$$

$$A_r = \frac{3}{2} A_g = \frac{3}{2} \cdot 8,59 = 12,89 \text{ mm}^2$$

$$A_s = \frac{4}{2} A_g = 2 \cdot 8,59 = 17,18 \text{ mm}^2$$

METAL FORMING

The height of an annealed carbon steel cylinder with a diameter equal to 50 mm and a height equal to 100 mm undergoes to open-die forging. The material is characterized by a flow curve with parameters $k = 500 \text{ MPa}$ and $n = 0,24$, and UTS equal to 350 MPa. The process requires two steps in both of which the height is reduced by 5%. Neglecting friction and barreling phenomenon, determine:

- The strains for each of the two steps and the overall strain.
- The ideal work for each of the two steps.

Solution

1. Determine the strains for each of the two steps and the overall strain

The process requires two steps in both of which the height is reduced by 5%, therefore the height of the cylinder will become, respectively:

$$\text{Step 1: } h_1 = h_0 \cdot (1 - 0,05) = 100 \cdot 0,95 = 95,00 \text{ mm}$$

$$\text{Step 2: } h_2 = h_1 \cdot (1 - 0,05) = 95 \cdot 0,95 = 90,25 \text{ mm}$$

Therefore, the true strains are:

$$\text{Step 1: } \varepsilon_1 = \varepsilon_{0-1} = \ln \frac{h_1}{h_0} = \ln \frac{95}{100} = -0,0513$$

$$\text{Step 2: } \varepsilon_{1-2} = \ln \frac{h_2}{h_1} = \ln \frac{90,25}{95} = -0,0513$$

$$\text{Overall: } \varepsilon_2 = \ln \frac{h_2}{h_0} = \ln \frac{90,25}{100} = -0,1026 = \varepsilon_{0-1} + \varepsilon_{1-2}$$

2. Determine the ideal work for each of the two steps

Please, recall that the strain energy density when dealing with the considered flow curve is equal to $u = \frac{k \cdot \varepsilon^{n+1}}{n+1}$, therefore:

$$\text{Step 1: } u_{0-1} = \frac{k \cdot \varepsilon_1^{n+1}}{n+1} = \frac{500 \cdot 0,0513^{0,24+1}}{0,24+1} = 10,141 \text{ MPa} = 0,010141 \frac{\text{J}}{\text{mm}^3}$$

$$\text{Overall: } u_{0-2} = \frac{k \cdot \varepsilon_2^{n+1}}{n+1} = \frac{500 \cdot 0,1026^{0,24+1}}{0,24+1} = 23,954 \text{ MPa} = 0,023954 \frac{\text{J}}{\text{mm}^3}$$

$$\text{Step 2: } u_{1-2} = u_{0-2} - u_{0-1} = 0,023954 - 0,010141 = 0,013813 \frac{\text{J}}{\text{mm}^3}$$

Now, considering the volume of the cylinder:

$$V = \frac{\pi}{4} d_o^2 \cdot h_0 = \frac{\pi}{4} 50^2 \cdot 100 = 196.350 \text{ mm}^3$$

The ideal work for each of the two steps will be:

$$\text{Step 1: } W_{0-1} = u_{0-1} \cdot V = 0,010141 \cdot 196.350 = 1.991 \text{ J} \cong 2 \text{ kJ}$$

$$\text{Step 2: } W_{1-2} = u_{1-2} \cdot V = 0,013813 \cdot 196.350 = 2.712 \text{ J} \cong 2,7 \text{ kJ}$$

MACHINING

A steel workpiece ($k_{cs} = 2800$ MPa, Kronenberg coefficient $x = 0,2$) must be machined on a planar surface with a length of 600 mm and a width of 20 mm. The operation involves a roughing step with a peripheral milling operation characterized by a machining allowance equal to 2 mm. The selected tool has a diameter of 20 mm, a working height of 30 mm and 4 straight cutting edges. The process parameters used are so that the feed velocity v_f is equal to 384 mm/min, and the number of revolutions per minutes n is equal to 1200 rpm.

Determine the chip thickness, the cutting force, the cutting torque and the cutting power, verifying that the operation can be performed with a milling machine with a power of 7,5 kW and characterized by an efficiency of 0,9.

Solution

In order to answer, it is necessary to understand what the actual situation in this slab milling operation is.

First of all, we need to compute the engagement angle and then the number of simultaneously working teeth:

$$\varphi = \arccos \frac{\frac{D}{2} - a_e}{\frac{D}{2}} = \arccos \frac{\frac{20}{2} - 2}{\frac{20}{2}} = \arccos 0,8 = 36,9^\circ = 0,644 \text{ rad}$$

$$\varphi_0 = \frac{2\pi}{Z} = \frac{2\pi}{4} = 1,571 \text{ rad} = 90^\circ$$

$$z = \frac{\varphi}{\varphi_0} = \frac{0,644}{1,571} = 0,41$$

Therefore, we have to consider the worst-case scenario when a single cutting edge is cutting the maximum chip thickness, that we have when the angular position of the cutting edge is equal to the engagement angle.

In order to do the computation, we need to determine the feed per tooth from the data.

$$n = 1.200 \text{ RPM}$$

$$v_f = 384 \frac{\text{mm}}{\text{min}}$$

$$v_f = Z \cdot f_z \cdot n \rightarrow f_z = \frac{v_f}{Z \cdot n} = \frac{384}{4 \cdot 1.200} = 0,08 \frac{\text{mm}}{\text{rev}}$$

Therefore:

$$h_{D,max} = h_D(\theta = \varphi) = f_z \sin \varphi = 0,08 \cdot \sin 36,9^\circ = 0,048 \text{ mm}$$

$$A_{D,max} = A_D(\theta = \varphi) = h_{D,max} \cdot a_p = 0,048 \cdot 20 = 0,96 \text{ mm}^2$$

$$k_{c,max} = k_{cs} h_{D,max}^{-x} = 2.800 \cdot 0,048^{-0,2} = 5.139,4 \text{ MPa}$$

$$F_{c,max} = k_{c,max} A_{D,max} = 5.139,4 \cdot 0,96 = 4.934 \text{ N}$$

$$T_c = F_{c,max} \frac{D/2}{1000} = 4.934 \frac{20/2}{1000} = 49,34 \text{ Nm}$$

$$\omega = \frac{2\pi n}{60} = \frac{2\pi \cdot 1.200}{60} = 125,67 \frac{\text{rad}}{\text{s}}$$

$$P_c = T_c \omega = 49,34 \cdot 125,67 = 6.200 \text{ W} = 6,2 \text{ kW}$$

$$P_{c,limit} = E \cdot P_e = 0,9 \cdot 7,5 = 6,75 \text{ kW}$$

Therefore, being $P_c = 6,2 \text{ kW} \leq P_{c,limit} = 6,75 \text{ kW}$, the slab milling operation is feasible.